Quiz #2 (Unit 2 - Extended): Floating-Point & Instruction Datapath

Total Marks: 160.00

IEEE 754 Conversion, Rounding Rules & ID/EX MUX Decoding

Name: **Student ID:** **Score:** / 160.00

> **Instructions:** Read the technique hint above each question. Adhere strictly to the designated question-type structure: Multiple Choice (Single Option), Select All That Apply (Multiple Options), or Fill in the Blank (Exact Values). Calculators are not permitted.

Question 1: Single-Precision Floating-Point Conversion

Mark 20.00 out of 20.00

[FILL IN THE BLANK - CALCULATION]

Technique / Concept Hint: Successive integer division by 2; successive fractional multiplication by 2; normalization shift E; biased exponent calculation ($E + 127$); mantissa zero-padding to 23 bits.

What is the single-precision (32-bit) IEEE 754 format of the decimal number **+29.625**? Fill in the blanks:

1. Sign bit: ____
2. Integer 29 in binary: ____
3. Fraction 0.625 in binary: ____
4. Normalized form: ____ × 2 ^ ____
5. Biased Exponent: 127 + ____ = ____ → ____ 2
6. Stored Mantissa (23 bits): ____

Question 2: Single-Precision Architecture Allocation

Mark 10.00 out of 10.00

[MULTIPLE CHOICE - SELECT ONE]

Technique / Concept Hint: IEEE 754 32-bit field width standards: exact bit allocations for sign, exponent, and fraction/mantissa.

In the IEEE 754 single-precision (32-bit) floating-point standard, how are the 32 bits allocated across the functional fields?

- ◯ a. 1 sign bit, 11 exponent bits, and 20 mantissa bits
- ◯ b. 1 sign bit, 8 exponent bits, and 23 mantissa bits
- ◯ c. 2 sign bits, 7 exponent bits, and 23 mantissa bits
- ◯ d. 1 sign bit, 7 exponent bits, and 24 mantissa bits

Question 3: Repeating Fraction with IEEE 754 Rounding

Mark 20.00 out of 20.00

[FILL IN THE BLANK - CALCULATION]

Technique / Concept Hint: Fractional multiplication loop detection; negative shift normalization ($E = -k$); IEEE rounding evaluation (Bit 24 = Guard, Bit 25 = Round, Bits 26+ = Sticky; round-up rule when Guard = 1 and Round/Sticky != 0).

Convert the repeating fraction **−1/6** to single-precision (32-bit) IEEE 754 format:

1. Sign bit: [ ]
2. Binary fraction of 1/6: 0. [ ] ... (show first 8 digits)
3. Normalized form: [ ] ... × 2 ^ [ ]
4. Biased Exponent: 127 + [ ] = [ ] → [ ] 2
5. Guard bit (24th bit) value: [ ] ; Sticky bit value: [ ]
6. Final 23-bit Mantissa after rounding: [ ]

Question 4: Hardware Rounding Logic & Status Flags

Mark 10.00 out of 10.00

[SELECT ALL THAT APPLY]

Technique / Concept Hint: IEEE 754 default round-to-nearest (ties to even); definitions and roles of the Guard bit, Round bit, and Sticky bit in floating-point ALUs.

Which of the following statements are **TRUE** regarding IEEE 754 hardware rounding logic? (Select all that apply.)

- [ ] a. If the Guard bit (the 24th bit in single precision) is 0, the remaining bits are simply truncated without rounding up.
- [ ] b. The Sticky bit is formed by taking the logical OR of all bits beyond the Round bit.
- [ ] c. When a tie occurs (Guard = 1, and all subsequent bits are 0), IEEE round-to-nearest ties to even forces the least significant bit (LSB) to 1.
- [ ] d. Rounding up is performed by adding 1 to the least significant bit (LSB) of the 23-bit mantissa.

Question 5: Double-Precision Floating-Point Conversion

Mark 20.00 out of 20.00

[FILL IN THE BLANK - CALCULATION]

Technique / Concept Hint: 64-bit IEEE 754 parameters: 1 sign bit, 11 exponent bits with bias 1023, 52 mantissa bits; calculation of trailing zeros to complete the 52-bit field.

Convert the decimal number **−6.375** to double-precision (64-bit) IEEE 754 format:

1. Sign bit: ____

2. Binary of integer 6: ____ ; Binary of fraction 0.375: ____

3. Normalized form: ____ × 2 ^ ____

4. Biased Exponent: 1023 + ____ = ____ → ____ 2 (11 bits)

5. Normalized fractional mantissa: ____

6. How many trailing zeros are needed after the fractional mantissa to complete the 52-bit field? ____

7. Final 64-bit representation in 4-bit nibbles:

```
Row 1 (Bits 63-32): ____
Row 2 (Bits 31-0): ____
```

Question 6: Double-Precision Exponent Characteristics

Mark 10.00 out of 10.00

[MULTIPLE CHOICE - SELECT ONE]

Technique / Concept Hint: Standard IEEE 754 64-bit double precision bias evaluation using formula $2^{11-1} - 1$.

What is the exact bias value added to the actual exponent in IEEE 754 double-precision (64-bit) floating-point format?

- ◯ a. 127
- ◯ b. 255
- ◯ c. 1023
- ◯ d. 2047

Question 7: Double-Precision Repeating Fraction & Rounding

Mark 20.00 out of 20.00

[FILL IN THE BLANK - CALCULATION]

Technique / Concept Hint: Multi-word repeating pattern expansion; 52-bit boundary tracking; 53rd bit guard inspection and tie-up rounding.

Convert the repeating decimal **+0.2** to double-precision (64-bit) IEEE 754 format:

1. Sign bit: [ ]

2. Repeating binary pattern of fraction 0.2: 0. [ ] ...

3. Normalized form: 1.100110011001... × 2 ^ [ ]

4. Biased Exponent: 1023 + [ ] = [ ] → [ ] 2

5. What are the **last 4 digits** of the 52-bit mantissa (including the LSB) after applying IEEE rounding rules?

[ ]

Question 8: Single vs. Double Precision Comparison

Mark 10.00 out of 10.00

[SELECT ALL THAT APPLY]

Technique / Concept Hint: Comparative analysis of single vs double precision: dynamic range, effective significand precision bits, bias magnitude, and storage trade-offs.

Which of the following statements accurately compare IEEE 754 single precision and double precision? (Select all that apply.)

- [ ] a. Double precision provides approximately 15 to 17 decimal digits of precision, whereas single precision provides approximately 7 to 8.
- [ ] b. Both formats dedicate exactly one bit to the sign.
- [ ] c. The dynamic range of double precision is expanded because its exponent field is 11 bits rather than 8 bits.
- [ ] d. Double precision stores the leading implicit bit in the register, while single precision omits it.

Question 9: ID/EX Pipeline Register Instruction & MUX Decoding

Mark 30.00 out of 30.00

[FILL IN THE BLANK - ARCHITECTURAL ANALYSIS]

Technique / Concept Hint: MIPS 32-bit instruction format parsing: Opcode [31-26], rs [25-21], rt [20-16], rd [15-11], funct [5-0], Immediate [15-0]; ALUSrc multiplexer input selection.

The ID/EX pipeline register holds instructions transitioning between the Decode and Execute stages:

```
Instruction A: 000000 10001 10010 01010 00000 100010
Instruction B: 100011 10000 01001 0000000000001100
```

Evaluate both instructions and the ALU source multiplexer actions:

1. **Instruction A:** Opcode (bits 31-26) is _____ 2 , classifying it as an _____ instruction.
   - The `rt` field spans bits _____ down to bit _____ , containing binary _____ .
   - The second ALU input MUX will select data from the: _____

(rt register / sign-extended immediate).

2. **Instruction B:** Opcode (bits 31-26) is _____ 2 , which corresponds to the MIPS _____

instruction (an I-Type instruction).

   - The 16-bit immediate field contains the binary pattern _____ , which equals decimal _____

.

   - For Instruction B, the ALU source MUX selects the: _____ to calculate the memory address.

Question 10: Datapath Multiplexer Control Generation

Mark 10.00 out of 10.00

[MULTIPLE CHOICE - SELECT ONE]

Technique / Concept Hint: Control unit decoding: ALUSrc control signal behavior across instruction types in the classic 5-stage RISC pipeline.

In the MIPS pipeline datapath, which control signal generated during Instruction Decode determines whether the second input to the ALU is taken from the register file (Read Data 2) or from the sign-extended immediate?

- ◯ a. RegDst
- ◯ b. ALUSrc
- ◯ c. MemtoReg
- ◯ d. Branch

Instructor Answer Key & Step-by-Step Solutions

Maximum Marks: 160.00

Quiz #2 (Unit 2 - Extended) - Verification & Marking Guide

Question 1 Solution: Converting +29.625 to Single Precision (20.00 Marks)

```
1. Sign bit: [0]
2. Integer 29 in binary: [11101]
3. Fraction 0.625 in binary: [101]
4. Normalized form: [1.1101101] × 2 ^ [4]
5. Biased Exponent: 127 + [4] = [131] → [10000011]_2
6. Stored Mantissa: [11011010000000000000000]
```

Verification: Integer 29 = 16 + 8 + 4 + 1 = 11101_2[cite: 9]. Fraction 0.625 = 0.5 + 0.125 = $.101_2$[cite: 9]. Normalized: 1.1101101×2^4[cite: 9]. Biased exponent: 4 + 127 = 131 = 10000011_2[cite: 2, 9]. Mantissa takes the fractional bits and pads with 16 zeros to make 23 bits[cite: 2, 9].

Question 2 Solution: Architecture Bitfields (10.00 Marks)

```
Correct Answer: [b]
```

Explanation: The IEEE 754 single-precision (32-bit) standard defines bit 31 as the sign bit (1 bit), bits 30-23 as the biased exponent (8 bits), and bits 22-0 as the fraction/mantissa (23 bits)[cite: 1, 9].

Question 3 Solution: Converting -1/6 with Rounding (20.00 Marks)

```
1. Sign bit: [1]
2. Binary fraction of 1/6: 0.[00101010]...
3. Normalized form: [1.01010101]... × 2 ^ [-3]
4. Biased Exponent: 127 + [-3] = [124] → [01111100]_2
5. Guard bit (24th bit): [1] ; Sticky bit: [1]
6. Final Mantissa after rounding: [01010101010101010101011]
```

Explanation: 1/6 in binary repeats: 0.00101010101... Shifting 3 places right yields $1.010101... \times 2^{-3}$[cite: 9]. Biased exponent is -3 + 127 = 124 (01111100_2)[cite: 2, 9]. The unrounded 23 bits are 01010101010101010101010[cite: 9]. Bit 24 (Guard) is 1, and the repeating sequence ensures Sticky = 1[cite: 9]. Under IEEE rounding rules, Guard = 1 and Sticky = 1 requires rounding up by adding 1 to the LSB, producing $...011$[cite: 9].

Question 4 Solution: Hardware Rounding Logic (10.00 Marks)

```
Correct Answers: [a, b, d]
```

Explanation:
• **a is TRUE:** If the Guard bit is 0, the discarded fraction is strictly less than half an LSB, so truncation is performed[cite: 9].
• **b is TRUE:** The sticky bit accumulates all bits below the round bit via logical OR[cite: 9].
• **c is FALSE:** In a tie (Guard = 1, Round = 0, Sticky = 0), round-to-nearest ties to even forces the LSB to **0** (even), not 1[cite: 9].
• **d is TRUE:** Rounding up increments the mantissa LSB by 1 using integer addition[cite: 9].

Question 5 Solution: Double Precision -6.375 (20.00 Marks)

```
1. Sign bit: [1]
2. Binary integer 6: [110] ; Binary fraction 0.375: [011]
3. Normalized form: [1.10011] × 2 ^ [2]
4. Biased Exponent: 1023 + [2] = [1025] → [10000000001]_2
5. Normalized fractional mantissa: [10011]
6. Trailing zeros needed: [47]
7. Row 1: [1100 0000 0001 1001 1000 0000 0000 0000]
   Row 2: [0000 0000 0000 0000 0000 0000 0000 0000]
```

Explanation: -6.375 is negative (Sign = 1)[cite: 9]. 6.375 = $\mathtt{110.011}_2$[cite: 9]. Normalization: 1.10011×2^2[cite: 9]. Double-precision bias is 1023, so Biased Exponent = 2 + 1023 = 1025 = $\mathtt{10000000001}_2$ (11 bits)[cite: 9]. The mantissa is 52 bits long; since `10011` occupies 5 bits, exactly 52 - 5 = 47 trailing zeros follow[cite: 9].

Question 6 Solution: Double-Precision Bias (10.00 Marks)

```
Correct Answer: [c]
```

Explanation: An 11-bit exponent field yields a bias of $2^{11-1} - 1 = 1024 - 1 = 1023$[cite: 9].

Question 7 Solution: Double Precision Repeating +0.2 (20.00 Marks)

```
1. Sign bit: [0]
2. Repeating binary pattern: 0.[001100110011]...
3. Normalized form: 1.100110011001... × 2 ^ [-3]
4. Biased Exponent: 1023 + [-3] = [1020] → [01111111100]_2
5. Last 4 digits with LSB after rounding: [1010]
```

Explanation: 0.2 in binary is 0.00110011... Normalized to $1.100110011... \times 2^{-3}$[cite: 2]. Biased Exponent = -3 + 1023 = 1020 = $\mathtt{01111111100}_2$[cite: 9]. The 52-bit mantissa consists of 13 copies of the 4-bit block `1001`[cite: 9]. The 53rd bit (Guard bit) is 1, and the subsequent repeating bits ensure Sticky = 1[cite: 9]. Under IEEE rounding rules, Guard = 1 and Sticky = 1 triggers a round-up: adding 1 to the final nibble `1001` yields `1010`[cite: 9].

Question 8 Solution: Single vs. Double Precision (10.00 Marks)

```
Correct Answers: [a, b, c]
```

Explanation:
• **a is TRUE:** Single precision has 24 effective bits (~7 decimal digits), double precision has 53 effective bits (~16 decimal digits).
• **b is TRUE:** Both formats allocate exactly 1 bit for sign (Bit 31 vs Bit 63)[cite: 1, 9].
• **c is TRUE:** Exponent increases from 8 bits to 11 bits, expanding the dynamic range from $10^{\pm 38}$ to $10^{\pm 308}$[cite: 9].
• **d is FALSE:** Both formats utilize an implicit leading bit for normalized numbers that is omitted from storage[cite: 2, 9].

Question 9 Solution: Pipeline ID/EX Register MUX Decoding (30.00 Marks)

```
1. Instruction A: Opcode = [000000] → [R-Type]
   • rt field spans bit [20] to bit [16], containing [10010]
   • Second ALU input MUX selects: [rt register]
2. Instruction B: Opcode = [100011] → [lw]
   • Immediate field contains [0000000000001100] = decimal [12]
   • Second ALU input MUX selects: [sign-extended immediate]
```

Explanation: Instruction A has opcode `000000` (R-Type, sub)[cite: 9]. The ALU operates on two register operands, so ALUSrc selects Read Data 2 from the `rt` register (bits 20-16: `10010`, register $s2)[cite: 9]. Instruction B has opcode `100011` (`lw`, load word)[cite: 9]. For memory reference instructions, the ALUSrc multiplexer selects the sign-extended 16-bit immediate (decimal +12) so the ALU can compute the effective data memory address ($s0 + 12)[cite: 7, 9].

Question 10 Solution: Datapath Multiplexer Control (10.00 Marks)

```
Correct Answer: [b]
```

Explanation: The `ALUSrc` control signal controls the 2-to-1 MUX feeding the second ALU operand: when `0`, it selects the `rt` register value; when `1`, it selects the sign-extended 16-bit immediate[cite: 9].
